LOCALLY RINGED LATTICES, THE DOUBLED-ORIGIN LINE, AND THE PROJECTIVE LINE

Abstract. We construct the affine spectrum of a commutative ring as a locally ringed distributive lattice. Localisation is characterised by its universal property, and the Zariski lattice by the universal property of support. Sections on an arbitrary lattice element are compatible families of elements of principal localisations. We prove finite gluing, an extension theorem for maps out of the structure sheaf, and the existence of its invertibility support. We identify the global sections of the affine spectrum with the original ring and prove its universal property for local morphisms. We define affineness using the canonical map to the spectrum of global sections and prove that it is equivalent to being isomorphic to an affine spectrum, and identify principal affine opens with localisation spectra. We construct the doubled-origin line, compute its global functions, prove its non-affineness, and classify its points by idempotents modulo the coordinate. Gluing with the reciprocal transition gives the projective line, a qcqs scheme over the base ring with only constant global functions, non-affine over a nonzero base. The last section records the formalisation status.

1. Conventions

All rings are commutative with identity; the zero ring is allowed, and homomorphisms preserve identity. A distributive lattice is bounded, with bottom $\bot$, top $\top$, meet $\wedge$, and join $\vee$. Lattice maps preserve both bounds and both binary operations. Empty joins are bottom.

The arguments are constructive. Existence assertions about representatives, denominators, exponents and finite presentations express mere existence; they do not specify choices. We use finite choice, which follows by induction for finite families, and unique choice. In particular, an object with a uniquely characterising property may be constructed using a merely existing presentation. No prime ideals or points of a spectrum are used in the constructions.

The ringed lattice approach to schemes was developed by Coquand, Lombardi and Schuster in [3]; their earlier paper [2] treats the projective spectrum as a distributive lattice. The constructive Zariski lattice and its algebraic order description are treated in Coquand and Lombardi [1]. We use the locally ringed lattice formulation of [4]. The affine construction is closely related to [5]; the elementary characterisation of localisation is also used in [6].

2. Localisation

Let S be a multiplicative submonoid of a ring A.

Definition 2.1. A homomorphism $f : A \to T$ is a *localisation at S* if $f(s)$ is a unit for every $s \in S$ and, for every homomorphism $g : A \to C$ inverting S, there is a unique homomorphism $h : T \to C$ such that $hf = g$.

The fraction construction gives such a localisation. Its elements are classes of pairs (a, s) with $s \in S$, where

$$(a, s) \sim (b, t) \quad \iff \quad \exists r \in S \; r(at - bs) = 0.$$

Addition and multiplication are given by the usual fraction formulas.

Proposition 2.2. *A homomorphism $f : A \to T$ is a localisation at S if and only if:*

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(1) *$f(s)$ is a unit for every $s \in S$;*
(2) *for every $x \in T$, there merely exist $a \in A$ and $s \in S$ with $f(s)x = f(a)$;*
(3) *if $f(a) = 0$, there merely exists $s \in S$ with $sa = 0$.*

Proof. Suppose the three conditions hold and $g : A \to C$ inverts S. First, $f(a) = f(b)$ implies $g(a) = g(b)$: some element of S annihilates $a - b$, and its image under g is a unit.

For $x \in T$, call $c \in C$ a value of x if, for every fraction equation $f(s)x = f(a)$, one has $g(s)c = g(a)$. Two fraction equations give $f(at) = f(bs)$, hence $g(at) = g(bs)$. Cancellation of units shows that the value calculated from either equation is the same. Thus x has a unique value, denoted $h(x)$. The fraction rules

$$\frac{a}{s} + \frac{b}{t} = \frac{at + bs}{st}, \qquad \frac{a}{s}\frac{b}{t} = \frac{ab}{st}$$

show that h is a ring homomorphism, and $hf = g$. Any other extension must give the same values, proving uniqueness.

The fraction construction satisfies the three conditions, the third by effectiveness of its defining equivalence relation. For any localisation $f : A \to T$, the universal properties give mutually inverse maps between T and the fraction ring, commuting with the maps from A. Transporting the three conditions along this isomorphism proves the converse. □

Corollary 2.3. *Two homomorphisms out of a localisation which agree on A are equal.*

Write A_g for a localisation at the powers of $g \in A$, and $f_g : A \to A_g$ for its structural map. The particular construction of A_g will not matter until a family of such rings is chosen.

3. The Zariski lattice

For a finite family $b = (b_1, \ldots, b_m)$, membership in the ideal (b) means mere existence of coefficients c_j with $x = \sum_j b_j c_j$. Define

$$x \in \sqrt{(b)} \quad \iff \quad \exists n \geq 0 \; x^n \in (b).$$

Lemma 3.1. *The radical of a finitely generated ideal is an ideal. Moreover, $x^n \in \sqrt{(b)}$ implies $x \in \sqrt{(b)}$.*

Proof. Closure under multiplication by a scalar follows from $(cx)^n = c^n x^n$. For addition use the elementary identity

$$(x + y)^{p+q} = Ux^p + Vy^q$$

for suitable $U, V \in A$. It is proved by double induction on p, q. If either exponent is zero the assertion is immediate. In the successor case, use the two inductive expressions for the common exponent $p + (q + 1) = (p + 1) + q$, multiply the first by x and the second by y, and add. This proves the identity without the binomial theorem. Finally, $(x^n)^m = x^{nm}$ proves the last assertion. □

For finite families put

$$a \preceq b \quad \iff \quad \text{every } a_i \text{ belongs to } \sqrt{(b)}.$$

This is a preorder. For transitivity, if $a \preceq b \preceq c$, a power of each a_i is a linear combination of the b_j. By Lemma 3.1 this combination belongs to $\sqrt{(c)}$, and so does a_i.

Definition 3.2. The *Zariski lattice* $\mathrm{Zar}(A)$ is the set of finite families modulo the equivalence $a \preceq b$ and $b \preceq a$. Write $[a]$ for a class and $D(x) = [(x)]$.

Join is concatenation of families; meet is the family of pairwise products. The empty family is bottom and (1) is top. Indeed, concatenation is a least upper bound for $\preceq$. Pairwise products give a greatest lower bound: if $x^p \in (a)$ and $x^q \in (b)$, then $x^{p+q} \in (a)(b)$. These facts also establish the lattice laws; distributivity follows by distributing pairwise products over concatenation.

Theorem 3.3 (Formal Nullstellensatz)**.** *For finite families a, b,*

$$[a] \le [b] \quad \Longleftrightarrow \quad a \preceq b.$$

Every element of $\mathrm{Zar}(A)$ *is merely a finite join of elements $D(x)$, and $[a] = \bigvee_i D(a_i)$.*

Proof. The order of the quotient lattice is characterised by $[a] \wedge [b] = [a]$. If $a \preceq b$, the greatest-lower-bound property identifies a with its family of products with b. Conversely, this identification implies $a \preceq ab \preceq b$. Finite generation follows from the description of join and the existence of representatives. □

Definition 3.4. A *support* of A in a distributive lattice M is a map $d : A \to M$ such that

$$d(0) = \bot, \quad d(1) = \top, \quad d(xy) = d(x) \wedge d(y), \quad d(x + y) \le d(x) \vee d(y).$$

Theorem 3.5 (Universal support)**.** *The map $D : A \to \mathrm{Zar}(A)$ is a support. Every support $d : A \to M$ factors uniquely through a lattice map $\bar{d} : \mathrm{Zar}(A) \to M$. Its value is*

$$\bar{d}([a]) = \bigvee_i d(a_i).$$

Proof. The support identities for D follow from ideal membership and the lattice operations. For a general support, induction gives

$$d(x) \le d(x^n), \qquad d\Big(\sum_j b_j c_j\Big) \le \bigvee_j d(b_j).$$

Thus $a \preceq b$ implies $\bigvee_i d(a_i) \le \bigvee_j d(b_j)$. The proposed formula descends to the quotient. It preserves joins by concatenation and meets by distributivity and $d(xy) = d(x) \wedge d(y)$; it also preserves both bounds. Uniqueness follows from finite generation. □

Proposition 3.6 (The localisation bridge)**.** *For $g, h \in A$,*

$$D(h) \le D(g) \quad \Longleftrightarrow \quad \exists n, y\ h^n = gy \quad \Longleftrightarrow \quad f_h(g) \in A_h^\times.$$

Proof. The first equivalence is Theorem 3.3. An equation $h^n = gy$ makes $f_h(g)$ a unit. Conversely, represent an inverse of $f_h(g)$ by y/h^n. Its inverse equation gives $f_h(gy - h^n) = 0$. Kernel detection gives $h^m(gy - h^n) = 0$ for some m, hence $h^{m+n} = g(h^m y)$. □

If $D(h) \le D(g)$, the universal property therefore gives a unique map

$$\rho_{h,g} : A_g \longrightarrow A_h, \qquad \rho_{h,g} f_g = f_h.$$

Corollary 2.3 gives identity and composition:

$$\rho_{g,g} = 1, \qquad \rho_{k,h}\rho_{h,g} = \rho_{k,g}.$$

These maps are independent of witnesses of the order relations.

4. Sections on the whole lattice

Choose a localisation A_g for every $g \in A$. For $u \in \mathrm{Zar}(A)$ define

$$\mathcal{O}_A(u) = \left\{ (s_g)_{D(g)\le u} \;\middle|\; s_g \in A_g, \quad \rho_{h,g}(s_g) = s_h \text{ if } D(h) \le D(g) \le u \right\}.$$

The index is the element g of the ring, not the lattice element $D(g)$. Thus no choice of a representative for a basic open is needed. When $D(g) = D(h)$, compatibility relates the components through the canonical mutually inverse restriction maps.

Ring operations are componentwise. Two sections are equal if all their components are equal. For $v \le u$, restriction forgets components; identity and composition follow from componentwise equality.

Proposition 4.1. *Evaluation at g is a natural ring isomorphism*

$$\epsilon_g : \mathcal{O}_A(D(g)) \xrightarrow{\cong} A_g.$$

Its inverse sends x to $(\rho_{h,g}(x))_{D(h)\le D(g)}$.

Proof. Composition of the ρ maps makes this family compatible. One inverse law is $\rho_{g,g} = 1$; the other is compatibility of sections. For $D(h) \le D(g)$, naturality is the same compatibility equation:

$$\epsilon_h(s|_{D(h)}) = \rho_{h,g}(\epsilon_g(s)).$$

□

Proposition 4.2 (Limit property)**.** *A ring map $T \to \mathcal{O}_A(u)$ is uniquely determined by compatible maps $T \to A_g$ for all $D(g) \le u$. Every such compatible family induces a map, by sending t to the family of its images.*

5. Finite gluing

Lemma 5.1 (Separation on a basic cover)**.** *Suppose $D(g) = \bigvee_{i=1}^n D(h_i)$. The maps $\rho_{h_i,g} : A_g \to A_{h_i}$ are jointly injective.*

Proof. It suffices to consider x restricting to zero. Write $f_g(s)x = f_g(y)$ with s a power of g. For each i, kernel detection gives a power t_i of h_i with $t_i y = 0$. Choose these finitely many witnesses. Since $D(h_i) \le D(t_i)$, the Nullstellensatz supplies $g^p = \sum_i t_i c_i$. Hence $yg^p = 0$, and cancellation of the units $f_g(g^p)$ and $f_g(s)$ gives $x = 0$. □

Theorem 5.2 (Basic sheaf lemma)**.** *Suppose $D(g) = \bigvee_i D(h_i)$ and $x_i \in A_{h_i}$ have equal images in $A_{h_i h_j}$ for every i, j. There is a unique $x \in A_g$ whose image in A_{h_i} is x_i for every i.*

Proof. Uniqueness is Lemma 5.1, so the type of amalgams is a proposition and merely existing fraction data may be used. Write $f_{h_i}(s_i)x_i = f_{h_i}(y_i)$, where s_i is a power of h_i. Agreement on overlaps implies that some power $(h_i h_j)^{m_{ij}}$ annihilates $y_i s_j - y_j s_i$. Choose all these finitely many exponents and a common bound M. Put

$$w_i = h_i^{M+1}, \qquad k_i = w_i s_i, \qquad z_i = w_i y_i.$$

Then $D(k_i) = D(h_i)$ and the cross relations hold in A itself:

$$z_i k_j = z_j k_i.$$

The positive exponent in w_i ensures $D(k_i) = D(h_i)$ even when the chosen denominator s_i is the zeroth power.

By the Nullstellensatz, merely $g^p = \sum_i k_i c_i$. Set $Y = \sum_i z_i c_i$. The cross relations give $Y k_j = z_j g^p$, so define

$$x = f_g(Y)\, f_g(g^p)^{-1}.$$

In A_{k_j}, cancellation of $f_{k_j}(g^p)$ gives $f_{k_j}(k_j)\rho_{k_j,g}(x) = f_{k_j}(z_j)$. The image of x_j satisfies the same equation, and $f_{k_j}(k_j)$ is a unit. Thus their images agree in A_{k_j}. Finally $A_{h_j} \to A_{k_j}$ is injective by Lemma 5.1, since their basic opens are equal. Therefore $\rho_{h_j,g}(x) = x_j$. The proof includes $n = 0$: in that case $D(g) = \bot$ and A_g is the zero ring. □

Definition 5.3. A presheaf of rings on a distributive lattice L consists of rings $\mathcal{O}(u)$ and restriction maps $\mathcal{O}(u) \to \mathcal{O}(v)$ for $v \le u$, satisfying identity and composition. It is a *sheaf* if every compatible family on a finite cover $u = \bigvee_i u_i$ has a unique amalgam in $\mathcal{O}(u)$. Compatibility means equality after restriction to $u_i \wedge u_j$. The empty cover is included, so $\mathcal{O}(\bot)$ is the terminal ring.

Theorem 5.4. *The presheaf $\mathcal{O}_A$ is a sheaf on* $\mathrm{Zar}(A)$.

Proof. First establish separation on arbitrary finite covers. If two sections agree on the basic generators of a finite presentation of v, their components at any $D(h) \le v$ agree: the opens $D(ha_j)$ cover $D(h)$, so apply Lemma 5.1. This extends agreement through binary and finite joins, including bottom, and proves separation.

Let $u = \bigvee_i u_i$ and let $s_i \in \mathcal{O}_A(u_i)$ be compatible. For $D(g) \le u$, consider sections $y \in \mathcal{O}_A(D(g))$ agreeing with every s_i on $D(g) \wedge u_i$. There is at most one such section, by separation on the cover $D(g) = \bigvee_i (D(g) \wedge u_i)$.

For existence, choose finite presentations of the u_i and concatenate them, retaining the chart index of each generator a_j. The opens $D(ga_j)$ cover $D(g)$, since

$$D(g) \wedge \bigvee_j D(a_j) = \bigvee_j D(ga_j).$$

Take the appropriate components of the s_i on these opens. Compatibility makes them agree on overlaps. Theorem 5.2 and Proposition 4.1 produce y; separation verifies its agreement with each s_i throughout $D(g) \wedge u_i$. Since y is uniquely characterised, this construction is independent of the chosen presentations.

Denote this section by y_g and set $s_g = \epsilon_g(y_g)$. If $D(h) \le D(g)$, the restriction of y_g is the uniquely characterised y_h. Naturality of evaluation gives $\rho_{h,g}(s_g) = s_h$. Thus the s_g form a section of $\mathcal{O}_A(u)$, and its restrictions are the s_i. Separation proves uniqueness. This argument applies unchanged to the empty cover. □

6. Extending maps from basic opens

Theorem 6.1. *Let G be a sheaf of rings on a distributive lattice M, let $\pi : \mathrm{Zar}(A) \to M$ be a lattice map, and suppose given ring maps*

$$\varphi_g : A_g \longrightarrow G(\pi D(g))$$

commuting with restriction. There is a unique family of ring maps

$$\Phi_u : \mathcal{O}_A(u) \longrightarrow G(\pi u)$$

commuting with restriction and satisfying $\Phi_{D(g)} = \varphi_g \epsilon_g$.

Proof. For $s \in \mathcal{O}_A(u)$, characterise $t \in G(\pi u)$ by

$$t|_{\pi D(g)} = \varphi_g(s_g) \qquad (D(g) \le u). \tag{1}$$

A finite presentation $u = \bigvee_j D(a_j)$ gives a cover $\pi u = \bigvee_j \pi D(a_j)$. Separation in G proves that at most one t satisfies (1).

The elements $\varphi_{a_j}(s_{a_j})$ are compatible: their restrictions to $\pi D(a_j a_k) = \pi D(a_j) \wedge \pi D(a_k)$ coincide by the assumed compatibility of φ and of s. Glue them to t. For an arbitrary $D(g) \le u$,

the opens $\pi D(ga_j)$ cover $\pi D(g)$. On each, both sides of (1) agree by restriction compatibility. Separation in G proves the equation. Thus t is independent of the presentation; put $\Phi_u(s) = t$.

The unit, sum and product of fitting elements fit the corresponding unit, sum and product of sections. Uniqueness therefore makes Φ_u a ring map. The same argument proves compatibility with restriction and the formula on basic opens. Any family with these properties satisfies (1), proving uniqueness. □

7. Invertibility opens and local structure

Definition 7.1. For a presheaf of rings $\mathcal{O}$ on L, an *invertibility open* of $s \in \mathcal{O}(u)$ is an element $v \le u$ such that, for all $w \le u$,

$$w \le v \quad \Longleftrightarrow \quad s|_w \in \mathcal{O}(w)^\times.$$

An *invertibility map* assigns such an open $D_u(s)$ to every section.

Lemma 7.2. *Invertibility opens are unique. If $v = D_u(s)$ and $w \le u$, then*

$$D_w(s|_w) = w \wedge v.$$

For every invertibility map,

$$D_u(1) = u, \qquad D_u(st) = D_u(s) \wedge D_u(t).$$

Proof. An invertibility open is the greatest open below u on which s is a unit, so it is unique. Restriction follows by testing on opens below w. The unit is invertible everywhere. In a commutative ring a product is a unit exactly when both factors are units, which proves the product formula by the same characterisation. □

Lemma 7.3. *In a sheaf of rings, being a unit is local for finite covers. Consequently the defining property of an invertibility open may be tested on a basis of finite covers. If $u = \bigvee_i u_i$ and the restrictions of s have invertibility opens $v_i \le u_i$, then $\bigvee_i v_i$ is the invertibility open of s.*

Proof. Local inverses agree on intersections by uniqueness of inverses, and therefore glue. The product of the glued inverse with the original section is one by separation. This also includes the empty cover. For the basis criterion, present any test open as a finite join of basic opens and apply this locality of units.

For the last assertion put $v = \bigvee_i v_i$. The section is a unit on each v_i, hence on v. Conversely, if it is a unit on $w \le u$, then $w \wedge u_i \le v_i$ for every i, and $w = \bigvee_i (w \wedge u_i) \le v$. □

Definition 7.4. A *locally ringed lattice* is a distributive lattice L with a sheaf of rings $\mathcal{O}$ and an invertibility map satisfying

$$D_u(0) = \bot, \qquad D_u(s+t) \le D_u(s) \vee D_u(t).$$

Together with Lemma 7.2, these say that $D_u : \mathcal{O}(u) \to {\downarrow}u$ is a support.

Theorem 7.5. *The ringed lattice $(\mathrm{Zar}(A), \mathcal{O}_A)$ is locally ringed. For a section $s \in \mathcal{O}_A(D(g))$ with $\epsilon_g(s) = a/g^n$,*

$$D_{D(g)}(s) = D(ga). \tag{2}$$

Proof. For $D(h) \le D(g)$, the fraction equation shows that the image of s in A_h is a unit exactly when $f_h(a)$ is a unit. By Proposition 3.6, this is equivalent to $D(h) \le D(a)$, hence to $D(h) \le D(ga)$. Evaluation is an isomorphism, so it preserves and reflects units. The basis criterion proves (2). Uniqueness makes the answer independent of the fraction. Finite presentations and Lemma 7.3 now give an invertibility open for every section on every u.

For zero, a basic open $D(b)$ where zero is a unit satisfies $D(b) \le D(0) = \bot$ by the bridge. Finite generation therefore gives $D_u(0) = \bot$.

For addition, test below $D_u(s+t)$ on a basic open $D(b)$ and write the evaluations of the restrictions of s, t as $a/p, c/q$ in A_b, with p, q powers of b. Since their sum is a unit, the identity

$$f_b(qa+pc) = f_b(p)f_b(q)\big(\epsilon_b(s|_{D(b)}) + \epsilon_b(t|_{D(b)})\big)$$

shows that $f_b(qa+pc)$ is a unit. The bridge and the support laws of D give

$$D(b) \le D(qa+pc) \le D(a) \vee D(c).$$

Intersecting with $D(b)$ gives $D(b) \le D(ba) \vee D(bc)$. By (2) and restriction compatibility these are contained respectively in $D_u(s)$ and $D_u(t)$. Finite generation proves the required addition inequality. □

We denote this locally ringed lattice by $\operatorname{Spec} A$.

A geometric morphism $X \to Y$ of ringed lattices consists of a lattice map $\pi : L_Y \to L_X$ and compatible ring maps $\pi^\sharp_u : \mathcal{O}_Y(u) \to \mathcal{O}_X(\pi u)$. It is *local* when

$$\pi(D^Y_u(s)) = D^X_{\pi u}(\pi^\sharp_u(s)).$$

The forward inequality always holds: ring maps preserve units and $\pi^\sharp$ commutes with restriction. Thus locality is precisely the reverse inequality. Notice the contravariance of the lattice map.

8. The universal property of the affine spectrum

Write $\Gamma X = \mathcal{O}_X(\top)$ for the ring of global sections of a locally ringed lattice X. For $a \in A$ and $u \in \operatorname{Zar}(A)$, the family $(f_g(a))_{D(g)\le u}$ is a section of $\mathcal{O}_A(u)$; denote it by $a|_u$. These sections commute with restriction and define ring maps $A \to \mathcal{O}_A(u)$.

Proposition 8.1. *The homomorphism*

$$\alpha_A : A \longrightarrow \Gamma(\operatorname{Spec} A), \qquad a \longmapsto a|_\top,$$

is an isomorphism. Moreover,

$$D_\top(a|_\top) = D(a).$$

Proof. Since $D(1) = \top$, evaluation identifies $\Gamma(\operatorname{Spec} A)$ with A_1. The identity of A inverts every power of 1, so the universal property of localisation supplies $b : A_1 \to A$ with $bf_1 = 1_A$. The maps $f_1 b$ and 1_{A_1} agree on A and hence are equal by Corollary 2.3. Thus f_1 is an isomorphism, and so is α_A. Explicitly, its inverse sends s to $b(\epsilon_1(s))$; the inverse laws follow from compatibility of the components of s. The support formula is (2) applied to $a/1$ on $D(1)$. □

Theorem 8.2 (Universal property of Spec)**.** *For any locally ringed lattice X and commutative ring A, local morphisms $X \to \operatorname{Spec} A$ correspond bijectively to ring homomorphisms $A \to \Gamma X$:*

$$\operatorname{Hom}_{\mathrm{loc}}(X, \operatorname{Spec} A) \cong \operatorname{Hom}_{\mathrm{Ring}}(A, \Gamma X).$$

The correspondence sends a morphism to its map on global sections, composed with α_A.

Proof. Let X have lattice M, and let $\varphi : A \to \mathcal{O}_X(\top)$ be a ring homomorphism. We construct a local morphism with inverse-image map $\pi : \operatorname{Zar}(A) \to M$ and maps $\pi^\sharp_u : \mathcal{O}_A(u) \to \mathcal{O}_X(\pi u)$.

The lattice map. The function

$$a \longmapsto D^X_\top(\varphi(a))$$

is a support, because φ is a ring map and $D^X_\top$ is a support. Theorem 3.5 supplies a unique lattice map π with

$$\pi D(g) = D^X_\top(\varphi(g)). \tag{3}$$

The maps on basic opens. Restrict φ to $\pi D(g)$. The defining property of the invertibility open says that the image of g is a unit there, and therefore so is the image of every power of g. The universal property of A_g gives a unique homomorphism

$$\varphi_g : A_g \longrightarrow \mathcal{O}_X(\pi D(g)), \qquad \varphi_g(f_g(a)) = \varphi(a)|_{\pi D(g)}.$$

For $D(h) \le D(g)$ the two maps

$$A_g \longrightarrow \mathcal{O}_X(\pi D(h))$$

obtained by restriction of φ_g and by $\varphi_h \rho_{h,g}$ agree on A. They are equal by Corollary 2.3.

The sheaf map. Theorem 6.1 extends these maps uniquely to a family $\pi^\sharp_u$ commuting with restriction, with

$$\pi^\sharp_{D(g)}(s) = \varphi_g(\epsilon_g(s)). \tag{4}$$

This defines a morphism of ringed lattices. On global sections it sends $a|_\top$ to $\varphi(a)$, by the defining equation of the extension at $D(1) = \top$.

Locality on basic opens. Suppose $\epsilon_g(s) = a/g^n$, put $w = \pi D(g)$, and write $t = \pi^\sharp_{D(g)}(s)$. The fraction equation gives

$$yt = \varphi(a)|_w, \qquad y = \varphi(g^n)|_w \in \mathcal{O}_X(w)^\times.$$

The product law for invertibility opens and restriction compatibility give

$$\begin{aligned} D^X_w(t) &= D^X_w(yt) = w \wedge D^X_\top(\varphi(a)) \\ &= \pi D(g) \wedge \pi D(a) = \pi D(ga) = \pi\big(D^{\operatorname{Spec} A}_{D(g)}(s)\big). \end{aligned}$$

Here the last equality is (2).

Locality on arbitrary opens. Present u as $\bigvee_i D(a_i)$. Both u and πu have the corresponding finite covers. Apply the join formula of Lemma 7.3 to s and $\pi^\sharp_u(s)$, using compatibility of $\pi^\sharp$ with restriction and the basic-open calculation. It follows that

$$D^X_{\pi u}(\pi^\sharp_u(s)) = \bigvee_i \pi\big(D^{\operatorname{Spec} A}_{D(a_i)}(s|_{D(a_i)})\big) = \pi\big(D^{\operatorname{Spec} A}_u(s)\big).$$

Thus the constructed morphism is local.

Uniqueness. Let a local morphism with lattice map ψ have global component φ. Proposition 8.1 and locality force

$$\psi D(a) = \psi\big(D^{\operatorname{Spec} A}_\top(a|_\top)\big) = D^X_\top(\varphi(a)).$$

Universal support gives $\psi = \pi$. With the lattice maps identified, consider its map on a basic open, composed with the inverse of ϵ_g. This is a map $A_g \to \mathcal{O}_X(\pi D(g))$ agreeing with φ_g on A, by restriction compatibility and the hypothesis on global sections. Corollary 2.3 identifies it with φ_g. The uniqueness clause of Theorem 6.1 then identifies all the sheaf maps with $\pi^\sharp$.

We have constructed a local morphism with prescribed global component and proved it unique. Consequently, taking the global component and constructing the morphism are mutually inverse operations. □

In particular, the identity homomorphism of ΓX determines a canonical local morphism

$$X \longrightarrow \mathrm{Spec}(\Gamma X).$$

Its construction uses the invertibility support of X; X need not already be given with affine charts.

9. Restriction, affineness and qcqs schemes

Proposition 9.1. *Let $X = (L, \mathcal{O}, D)$ be a locally ringed lattice and $e \in L$. There is a locally ringed lattice $X|_e$ on*

$$\downarrow e = \{u \in L \mid u \leq e\},$$

with section ring $\mathcal{O}(u)$ at $u \leq e$ and the inherited restriction maps. Its invertibility support is the support of X, viewed in this principal ideal. The inclusion $X|_e \to X$ is local.

Proof. Meets, joins and bottom are inherited from L, and top is e. A finite cover in $\downarrow e$ is a finite cover of the same element in L, so the sheaf condition is inherited. For $s \in \mathcal{O}(u)$, the open $D_u(s)$ already lies below $u \leq e$. Testing it on opens below u gives exactly the same invertibility characterisation as in X. Its support laws are inherited as well.

The inverse-image map of the inclusion is $u \mapsto u \wedge e$, a lattice map from L to $\downarrow e$. Its maps on rings are restriction $\mathcal{O}(u) \to \mathcal{O}(u \wedge e)$. Their naturality is composition of restrictions. Locality is the identity

$$D_{u\wedge e}(s|_{u\wedge e}) = D_u(s) \wedge e,$$

which follows from Lemma 7.2 and $D_u(s) \leq u$. □

Proposition 9.2 (Universal property of restriction)**.** *A local morphism $f : Z \to X$ factors through $X|_e \to X$ if and only if $f^*(e) = \top$. The factorisation is unique.*

Proof. If $f^*(e) = \top$, restrict its lattice map to $\downarrow e$ and retain its section maps there. This is a lattice map because the new top e maps to top; the other lattice laws and locality are inherited. Composing with the inclusion recovers f: on lattices, $f^*(u \wedge e) = f^*(u)$, and on rings the equality is naturality with respect to restriction. Conversely, the inverse image of e under the inclusion is top in $\downarrow e$. Uniqueness follows because every $u \leq e$ satisfies $u \wedge e = u$, and restriction between these equal opens is the identity. □

An isomorphism of ringed lattices may be characterised by a bijective inverse-image map on lattices and bijective maps on all the section rings. The inverse lattice map and inverse ring maps supply its inverse morphism; naturality follows from that of the original maps.

Lemma 9.3. *Every isomorphism of ringed lattices between locally ringed lattices is local.*

Proof. Write its lattice map as $\pi : L_Y \to L_X$. Given $s \in \mathcal{O}_Y(u)$, put $v = D^X_{\pi u}(\pi^\sharp_u(s))$ and let $w = \pi^{-1}(v)$. Since $v \leq \pi u$ and π is a lattice isomorphism, $w \leq u$. Naturality shows that the image of $s|_w$ under the section-ring isomorphism at w is a unit. Ring isomorphisms reflect units, so $s|_w$ is a unit. Hence $w \leq D^Y_u(s)$ and $v \leq \pi D^Y_u(s)$. The opposite inequality holds for every morphism of ringed lattices, proving locality. □

Identities and composites of local morphisms are local, directly from the defining support equation. Isomorphisms compose. Restriction between equal opens is a ring isomorphism, since the two restriction maps are inverse by the restriction laws. Consequently, identifying two lattice maps pointwise and identifying their section maps through these restriction isomorphisms preserves the property of being an isomorphism.

Proposition 9.4. *A ring homomorphism $\beta : B \to A$ induces a local morphism*

$$\operatorname{Spec}\beta : \operatorname{Spec} A \longrightarrow \operatorname{Spec} B.$$

It is an isomorphism if β is a ring isomorphism.

Proof. Apply Theorem 8.2 to $\alpha_A\beta : B \to \Gamma(\operatorname{Spec} A)$. If $\psi : X \to \operatorname{Spec} A$ has global component $\varphi : A \to \Gamma X$, the composite $(\operatorname{Spec}\beta)\psi$ has global component $\varphi\beta$, by composition of the maps on global sections.

If $\gamma : A \to B$ is inverse to β, the two composites of $\operatorname{Spec}\beta$ and $\operatorname{Spec}\gamma$ have the same global components as the respective identity morphisms. Uniqueness in Theorem 8.2 identifies them with those identities. Thus the two induced morphisms are inverse isomorphisms. □

Definition 9.5. A locally ringed lattice X is *affine* if its canonical morphism

$$c_X : X \longrightarrow \operatorname{Spec}(\Gamma X),$$

corresponding to the identity of ΓX, is an isomorphism.

Theorem 9.6 (Affine criterion)**.** *A locally ringed lattice is affine if and only if it is isomorphic to* $\operatorname{Spec} A$ *for some commutative ring A.*

Proof. If X is affine, take $A = \Gamma X$. Conversely, let $\psi : X \xrightarrow{\sim} \operatorname{Spec} A$ be an isomorphism. It is local by Lemma 9.3. Its global component is a ring isomorphism

$$\alpha : A \xrightarrow{\alpha_A} \Gamma(\operatorname{Spec} A) \xrightarrow{\Gamma\psi} \Gamma X.$$

The second arrow is an isomorphism because the section map at top is bijective and restriction between the equal top opens is bijective. Proposition 9.4 gives the isomorphism

$$\operatorname{Spec}(\alpha^{-1}) : \operatorname{Spec} A \xrightarrow{\sim} \operatorname{Spec}(\Gamma X).$$

The composite $\operatorname{Spec}(\alpha^{-1})\psi$ is local and has global component $\alpha\alpha^{-1} = 1_{\Gamma X}$. By Theorem 8.2 it equals c_X. Therefore c_X is an isomorphism. □

Definition 9.7. An open u of a locally ringed lattice X is *affine* if $X|_u$ is affine. An *affine cover* is a finite family of affine opens $u_1, \ldots, u_n$ with $\bigvee_i u_i = \top$. A *qcqs scheme*, in the lattice formulation, is a locally ringed lattice for which an affine cover merely exists.

The affine cover is a witness of a property, not part of the structure. Its intersections are not required to be affine. In particular, two affine opens with join top suffice to exhibit a qcqs scheme. The finite cover may be empty.

Theorem 9.8 (Principal affine opens)**.** *For $g \in A$, there is an isomorphism of locally ringed lattices*

$$(\operatorname{Spec} A)|_{D(g)} \cong \operatorname{Spec}(A_g).$$

In particular, every principal open of an affine spectrum is affine.

Proof. Write $\beta : A \to A_g$ for localisation. The morphism $\operatorname{Spec}\beta : \operatorname{Spec}(A_g) \to \operatorname{Spec} A$ sends $D(g)$ to top, since $\beta(g)$ is a unit. Proposition 9.2 therefore gives

$$j : \operatorname{Spec}(A_g) \longrightarrow (\operatorname{Spec} A)|_{D(g)}.$$

The inverse of evaluation in Proposition 4.1 is a ring map

$$A_g \longrightarrow \mathcal{O}_A(D(g)) = \Gamma((\operatorname{Spec} A)|_{D(g)}).$$

Theorem 8.2 gives a morphism k in the other direction. To show $kj = 1$, compare their global components. It suffices to check them on $\beta(a)$ by uniqueness for localisation; both give the global section of $\beta(a)$. The universal property of Spec then proves equality of the morphisms.

For $jk = 1$, compose both sides with the inclusion into $\operatorname{Spec} A$. Their global components agree on every $a \in A$, so Theorem 8.2 gives equality after inclusion. Uniqueness in Proposition 9.2 cancels the inclusion. Thus j and k are inverse. Affineness follows from Theorem 9.6. □

10. Doubling along an open

Let $X = (L, \mathcal{O}, D)$ be a locally ringed lattice and let $d \in L$.

Definition 10.1. The *doubled lattice along* d is

$$L_2 = \{(u, v) \in L \times L \mid u \wedge d = v \wedge d\}.$$

For $x = (u, v)$ write $m(x) = u \wedge d = v \wedge d$.

Both bounds and both operations are componentwise. Distributivity shows that the operations preserve the defining equation. Order is componentwise, and the two projections preserve finite joins and meets.

Recall the ring fibre product

$$A \times_C B = \{(a, b) \mid f(a) = g(b)\}$$

for maps $f : A \to C$, $g : B \to C$. It is characterised by the fact that two maps into A, B with equal composites into C induce a unique map into $A \times_C B$. A pair is a unit if and only if its components are: the images of their inverses coincide by uniqueness of inverses in C.

Define a presheaf on L_2 by

$$\mathcal{O}_2(u, v) = \mathcal{O}(u) \times_{\mathcal{O}(m(u,v))} \mathcal{O}(v).$$

Restrictions are componentwise. The universal property of the fibre product constructs these ring maps; their identity and composition laws follow by the two projections.

Theorem 10.2. *The presheaf $\mathcal{O}_2$ is a sheaf, and*

$$D^2_{(u,v)}(s, t) = \big(D_u(s), D_v(t)\big)$$

is an invertibility support. Thus $(L_2, \mathcal{O}_2, D^2)$ is a locally ringed lattice.

Proof. For a compatible family on a finite cover of (u, v), glue its first components in $\mathcal{O}(u)$ and its second components in $\mathcal{O}(v)$, giving s, t. Their restrictions to $m(u, v)$ agree: the corresponding overlap opens of the covering family cover $m(u, v)$, by distributivity, and the restrictions agree on each by the fibre-product conditions. Separation gives agreement on all of $m(u, v)$. Thus (s, t) is an amalgam; its uniqueness follows componentwise. The argument includes the empty cover.

The proposed invertibility open belongs to L_2. Indeed, since $D_u(s) \le u$,

$$D_u(s) \wedge d = m(u, v) \wedge D_u(s) = D_{m(u,v)}(s|_{m(u,v)}).$$

The same expression for t is equal, since the overlap sections agree. For $(w, z) \le (u, v)$, the restriction of (s, t) is a unit if and only if both component restrictions are units, equivalently $w \le D_u(s)$ and $z \le D_v(t)$. This proves the invertibility characterisation. The zero and addition support laws hold componentwise. □

Proposition 10.3 (The two charts)**.** *Put $e_0 = (\top, d)$ and $e_1 = (d, \top)$. Then*

$$e_0 \vee e_1 = \top, \qquad e_0 \wedge e_1 = (d, d).$$

The original locally ringed lattice is isomorphic to each restriction $X_2|_{e_i}$, by

$$\iota_0(u) = (u, u \wedge d), \qquad \iota_1(u) = (u \wedge d, u).$$

More precisely, ι_i identifies L with the opens below e_i, and there are natural ring isomorphisms

$$\mathcal{O}(u) \cong \mathcal{O}_2(\iota_i(u))$$

carrying invertibility opens to invertibility opens.

Proof. If $(a, b) \le e_0$, then $b \le d$ and $b = b \wedge d = a \wedge d$. Thus $(a, b) = \iota_0(a)$. This proves the first lattice identification, and the second is symmetric. Under the first, the section isomorphism is

$$s \longmapsto (s, s|_{u\wedge d}),$$

with inverse first projection. Its inverse laws follow from the overlap condition. It commutes with restriction, and

$$D^2_{\iota_0(u)}(s, s|_{u\wedge d}) = (D_u(s), D_u(s) \wedge d) = \iota_0(D_u(s)).$$

For the second chart use $s \mapsto (s|_{u\wedge d}, s)$ and second projection. Its overlap is $(u \wedge d) \wedge d$, equal to $u \wedge d$; the restriction identities supply the same inverse laws. □

10.1. **Maps into a doubling.** There is a local *fold morphism* $p : X_2 \to X$. Its lattice map is $a \mapsto (a, a)$, and its section maps are $s \mapsto (s, s)$. Its locality follows from the componentwise invertibility support.

Theorem 10.4. *For any locally ringed lattice Z, local morphisms $Z \to X_2$ correspond to triples (f, u_0, u_1) with*

$$f : Z \to X \text{ local}, \qquad u_0 \vee u_1 = \top, \qquad u_0 \wedge u_1 = f^*(d).$$

The correspondence takes ψ to $(p\psi, \psi^* e_0, \psi^* e_1)$. Thus the extents are the full inverse images of the two charts.

Proof. For a triple as stated, define the inverse-image map by

$$\pi(a, b) = (u_0 \wedge f^* a) \vee (u_1 \wedge f^* b).$$

It preserves bounds and joins. For meets, expand into four terms. The two mixed terms lie below $u_0 \wedge u_1 = f^* d$, and $a \wedge d = b \wedge d$ identifies the two coordinates there. Each mixed term therefore lies below the required unmixed terms. This proves preservation of meets.

For $(s, t) \in \mathcal{O}_2(a, b)$, apply the section maps of f and restrict to $u_0 \wedge f^* a$ and $u_1 \wedge f^* b$. On their intersection the images agree, by the defining overlap equation of (s, t). Glue in Z. Uniqueness of the amalgam proves preservation of ring operations and compatibility with restriction. The join formula for invertibility opens, and locality of f, prove locality of the constructed map.

Its composite with the fold has lattice map f^* and the same section maps as f, by separation. Its chart inverse images are u_0, u_1, using $f^* d \le u_0, u_1$.

Conversely every $(a, b) \in L_2$ decomposes as

$$(a, b) = \big(e_0 \wedge (a, a)\big) \vee \big(e_1 \wedge (b, b)\big).$$

Indeed, its first coordinate is $a \vee (d \wedge b) = a$ and its second is $(d \wedge a) \vee b = b$. Applying a lattice map to this identity recovers the displayed formula for π. The section map is recovered by separation on the same two pieces. These give both inverse laws. □

11. The line with doubled origin

Let k be any commutative ring, put $A = k[X]$, and take $d = D_A(X)$. Apply Theorem 10.2 to $\operatorname{Spec} A$. The resulting locally ringed lattice has compact opens

$$\left\{(u, v) \in \operatorname{Zar}(k[X])^2 \;\middle|\; u \wedge D(X) = v \wedge D(X)\right\}$$

and section rings

$$\mathcal{O}_2(u, v) = \mathcal{O}_A(u) \times_{\mathcal{O}_A(u \wedge D(X))} \mathcal{O}_A(v).$$

By Proposition 10.3, its two chart opens $(\top, D(X))$ and $(D(X), \top)$ cover it, and each chart is a copy of $\operatorname{Spec} k[X]$. Their intersection corresponds in either chart to $D(X)$, whose section ring is $k[X, X^{-1}]$ by Proposition 4.1. The transition identifies the two coordinates, rather than inverting them. This is the line with doubled origin: two copies of the affine line identified along the punctured line. The construction permits an arbitrary base ring, including the zero ring.

Theorem 11.1. *The line with doubled origin over any commutative ring k is a qcqs scheme.*

Proof. Each chart restriction is isomorphic to $\operatorname{Spec} k[X]$ by Proposition 10.3, now regarded as an isomorphism with the restriction of Proposition 9.1. Both restrictions are affine by Theorem 9.6. The chart opens $e_0 = (\top, D(X))$ and $e_1 = (D(X), \top)$ satisfy

$$e_0 \vee e_1 = (\top \vee D(X), D(X) \vee \top) = (\top, \top).$$

They therefore form a two-element affine cover, giving the property in Definition 9.7. No nontriviality hypothesis on k is needed. □

11.1. **Global sections.** Continue to write $A = k[X]$, and denote the doubled-origin line by S.

Lemma 11.2. *The localisation homomorphism $A \to A_X$ is injective. Consequently, global sections of* $\operatorname{Spec} A$ *which agree on $D(X)$ are equal.*

Proof. Multiplication by X is injective on $k[X]$: equality of coefficients in $Xp = Xq$ gives equality of the coefficients of p and q. Induction shows that multiplication by every power X^n is injective. If a, b have the same image in A_X, kernel detection for localisation gives some n with $X^n(a - b) = 0$. Hence $a = b$.

By Proposition 8.1, write the two global sections as $a|_\top, b|_\top$. Their components on $D(X)$ are $f_X(a), f_X(b)$. Agreement on $D(X)$ therefore implies $a = b$, proving the second claim. □

Proposition 11.3. *Every global section (s, t) of S has $s = t$. The diagonal homomorphism*

$$\delta : k[X] \longrightarrow \Gamma S, \qquad a \longmapsto (a|_\top, a|_\top),$$

is a ring isomorphism.

Proof. The two components of a global section agree on $\top \wedge D(X) = D(X)$ by the fibre-product condition. They are equal by Lemma 11.2. The diagonal map is a ring map by the fibre-product universal property. An inverse sends (s, t) to the element of $k[X]$ corresponding to its first component under Proposition 8.1. One inverse law is the corresponding law for $\Gamma(\operatorname{Spec} A) \cong A$; the other uses $s = t$. □

Equivalently, under the affine global-section identifications,

$$\Gamma S \cong k[X] \times_{k[X, X^{-1}]} k[X] \cong k[X].$$

The two chart restrictions of a global function thus give the same polynomial. No assumption that k is a domain is involved.

11.2. **Non-affineness.**

Theorem 11.4. *If the doubled-origin line over k is affine, then $1 = 0$ in k. In particular, if $1 \neq 0$ in k, it is a non-affine qcqs scheme.*

Proof. Let $\pi : \mathrm{Zar}(\Gamma S) \to L_2$ be the inverse-image lattice map of the canonical morphism $c_S : S \to \mathrm{Spec}(\Gamma S)$. Call $(u, v) \in L_2$ *diagonal* when $u = v$. Diagonal opens are closed under finite joins, including the empty join, since the lattice operations are componentwise.

For a global section $r = (s, t)$, Proposition 11.3 gives $s = t$. Its invertibility open is consequently diagonal:

$$D^S_\top(r) = (D^A_\top(s), D^A_\top(t)).$$

By the construction of the canonical morphism,

$$\pi D_{\Gamma S}(r) = D^S_\top(r).$$

Every element of $\mathrm{Zar}(\Gamma S)$ merely has a finite presentation by generators $D_{\Gamma S}(r_i)$. Since π preserves finite joins, every open in its image is diagonal. This argument does not require an identification of $\mathrm{Zar}(\Gamma S)$ with $\mathrm{Zar}(k[X])$.

If S is affine, c_S is an isomorphism, so π is surjective. In particular the first chart $e_0 = (\top, D_A(X))$ lies in its image. It must therefore be diagonal, giving $D_A(1) = \top = D_A(X)$. The localisation bridge, or the formal Nullstellensatz, now supplies n and $q \in k[X]$ with

$$1 = 1^n = Xq.$$

Apply the ring homomorphism $k[X] \to k$ evaluating X at zero. It gives $1 = 0\, q(0) = 0$ in k, as required. □

11.3. **Points and their extents.** The inclusion $k \to k[X]$ followed by δ of Proposition 11.3 defines a morphism $S \to \mathrm{Spec}\, k$ by Theorem 8.2. For a k-algebra B, a B-valued point over k means a local morphism $\mathrm{Spec}\, B \to S$ whose composite to $\mathrm{Spec}\, k$ is induced by the given structure map $k \to B$.

Theorem 11.5. *For every k-algebra B there is an equivalence*

$$\mathrm{Hom}_k(\mathrm{Spec}\, B, S) \cong \left\{ (t, u_0, u_1) \;\middle|\; \begin{array}{ll} t \in B, & u_0, u_1 \in \mathrm{Zar}(B), \\ u_0 \vee u_1 = \top, & u_0 \wedge u_1 = D_B(t) \end{array} \right\}.$$

Here u_i is the full inverse image of the ith chart. For a k-algebra map $h : B \to B'$, precomposition with $\mathrm{Spec}\, h$ *sends the coordinates to*

$$(h(t), \mathrm{Zar}(h)(u_0), \mathrm{Zar}(h)(u_1)),$$

where $\mathrm{Zar}(h)(D_B(b)) = D_{B'}(h(b))$.

Proof. Apply Theorem 10.4 with $Z = \mathrm{Spec}\, B$ and $X = \mathrm{Spec}\, k[X]$. A map $f : \mathrm{Spec}\, B \to \mathrm{Spec}\, k[X]$ over k is, by Theorem 8.2 and Proposition 8.1, a k-algebra homomorphism $k[X] \to B$. The polynomial universal property identifies it with the image t of X. Locality gives $f^* D(X) = D_B(t)$. This identifies the triples and proves the equivalence.

For base change, the global component of the composite is the composite algebra homomorphism, so the coordinate becomes $h(t)$. The inverse images of the chart opens compose, giving the two asserted extents. The generator formula for $\mathrm{Zar}(h)$ follows from the support equation for the spectrum morphism and Proposition 8.1. □

11.4. **Complementary opens and idempotents.** Write $\mathrm{Idem}(C) = \{e \in C \mid e^2 = e\}$. The next two results explain how the pair of chart extents becomes an idempotent modulo t.

Proposition 11.6. *In any locally ringed lattice X, idempotents of ΓX correspond to complementary pairs of opens. The correspondence is*

$$e \longmapsto \big(D_\top(e), D_\top(1-e)\big).$$

Proof. For $e^2 = e$, the identities $e(1-e) = 0$ and $e + (1-e) = 1$ show that the two supports have meet bottom and join top.

Conversely let $u \vee v = \top$ and $u \wedge v = \bot$. Glue 1 on u with 0 on v. They agree on the intersection because the ring on bottom is terminal, by the empty-cover sheaf condition. Denote the amalgam by e. Separation gives $e^2 = e$.

More generally, a section with restrictions 1 on u and 0 on v has support u: restriction compatibility gives $u \wedge D_\top(e) = u$ and $v \wedge D_\top(e) = \bot$, and distributivity over $u \vee v = \top$ gives $D_\top(e) = u$. The same argument gives $D_\top(1-e) = v$.

For the other inverse law, an idempotent unit is one, so $e|_{D_\top(e)} = 1$. On $D_\top(1-e)$ the element $1-e$ is a unit annihilating e, so the restriction of e is zero. Separation therefore recovers e from its complementary pair. □

Proposition 11.7. *For $t \in B$, there is a canonical lattice isomorphism*

$$\mathrm{Zar}(B/(t)) \cong [D_B(t), \top].$$

If $q : B \to B/(t)$ is the quotient map, the image of $D(q(b))$ is $D_B(t) \vee D_B(b)$. The interval has bottom $D_B(t)$.

Proof. Both lattices have the universal property of a support of B killing t. For the interval use the support $b \mapsto D_B(t) \vee D_B(b)$. A support $d : B \to N$ with $d(t) = \bot$ extends uniquely to $\mathrm{Zar}(B)$; restriction of that lattice map to the interval preserves bottom and has the desired factorisation. Uniqueness follows by precomposing with $u \mapsto D_B(t) \vee u$, which is onto the interval.

For the quotient, a support killing t descends uniquely to $B/(t)$. Indeed, if $b' = b + tc$, then $d(b') \leq d(b) \vee d(tc) = d(b)$, and the reverse inequality follows in the same way. The support laws descend on representatives. Universal support for $B/(t)$ now gives the second universal property. The two resulting lattice maps are inverse by uniqueness. Their values on $q(b)$ give the formula. □

Theorem 11.8. *For every k-algebra B, naturally in B,*

$$\mathrm{Hom}_k(\mathrm{Spec}\, B, S) \cong \sum_{t \in B} \mathrm{Idem}(B/(t)).$$

Proof. For fixed t, the pairs of Theorem 11.5 are precisely the complementary pairs in the interval $[D_B(t), \top]$. Proposition 11.7, Proposition 11.6 applied to $\mathrm{Spec}(B/(t))$, and Proposition 8.1 identify them with $\mathrm{Idem}(B/(t))$.

Explicitly, if an idempotent is represented by $q(b)$, its extents are

$$u_0 = D_B(t) \vee D_B(b), \qquad u_1 = D_B(t) \vee D_B(1-b).$$

This is independent of the representative by the quotient-lattice isomorphism. A map $h : B \to B'$ induces $\bar{h} : B/(t) \to B'/(h(t))$, and the formula commutes with the action of $\mathrm{Zar}(h)$. Thus base change sends (t, e) to $(h(t), \bar{h}(e))$. The formula includes the zero ring: if t is a unit, $B/(t)$ has just one idempotent and the extents are $(\top, \top)$. □

12. Gluing with a transition

Let X_1, X_2, Y be locally ringed lattices and let $a_i : Y \to X_i$ be local morphisms, in geometric notation. Write $p_i : L_i \to M$ for their inverse-image lattice maps. There is a componentwise construction on the pullback lattice

$$L = \{(u, v) \in L_1 \times L_2 \mid p_1(u) = p_2(v)\}.$$

At (u, v), put $w = p_1(u) = p_2(v)$ and define

$$\mathcal{O}(u, v) = \mathcal{O}_1(u) \times_{\mathcal{O}_Y(w)} \mathcal{O}_2(v),$$

using the two section maps of a_i. Operations and restrictions are componentwise. As in Theorem 10.2, compatible sections glue in X_1 and X_2, and separation in Y verifies the overlap equation for the amalgams. Locality of a_i ensures that

$$D_{(u,v)}(s, t) = (D^1_u(s), D^2_v(t))$$

belongs to the pullback lattice and is the invertibility open. The support laws hold componentwise. This construction gives a locally ringed lattice for any such span; the chart statements below use the particular open inclusions and transition isomorphism.

Now take $X_1 = X_2 = X$, $Y = X|_d$, and let $\theta : X|_d \xrightarrow{\sim} X|_d$ be an automorphism. Take a_1 to be the inclusion and a_2 the inclusion after θ. Write the result as X^θ. Its opens and sections are

$$L_\theta = \{(u, v) \mid u \wedge d = \theta^*(v \wedge d)\},$$
$$\mathcal{O}_\theta(u, v) = \{(s, t) \mid s|_{u\wedge d} = \theta^\sharp_{v\wedge d}(t|_{v\wedge d})\}.$$

Equal opens in the section equation are identified. For the identity transition this is the doubling construction.

Proposition 12.1. *The opens $e_0 = (\top, d)$ and $e_1 = (d, \top)$ cover X^θ, and both chart restrictions are isomorphic to X.*

Proof. An automorphism of $\downarrow d$ preserves its top d, so the chart pairs belong to L_θ, and their join is top. Below e_0, the second component lies below d and is uniquely determined by the first:

$$v = (\theta^*)^{-1}(u \wedge d).$$

The first projection identifies the chart lattice with L_X. On sections the second leg into the overlap ring is an isomorphism, so first projection from the fibre product is an isomorphism. These projections commute with restriction and hence give a ringed-lattice isomorphism; it is local by Lemma 9.3.

Below e_1, one has $u = \theta^*(v \wedge d)$. Use second projection instead: the first leg into the overlap ring is restriction between equal opens and hence an isomorphism. This proves the second chart identification. The second argument in fact needs no invertibility of θ. □

The construction and chart identifications do not assert a general mapping property for gluing with a transition. Unlike the identity doubling, a nontrivial transition need not admit a coordinate fold back to X.

13. The projective line

13.1. **Inversion on the punctured chart.** Put $A = k[X]$, $L = A_X$, $f : A \to L$, and $x = f(X)$. Let x^{-1} be its inverse. The polynomial universal property gives a homomorphism $\tau : A \to L$ fixing the image of k and sending X to x^{-1}. Since $\tau(X)$ is a unit, localisation extends it uniquely to

$$\sigma : L \longrightarrow L, \qquad \sigma f = \tau.$$

It fixes k and satisfies $\sigma(x) = x^{-1}$. Ring maps preserve inverses, so $\sigma(x^{-1}) = x$. Thus $\sigma^2 f$ and f agree on k and on X, and are equal by the polynomial universal property. Corollary 2.3 gives $\sigma^2 = 1_L$.

Apply Spec and use Theorem 9.8 to obtain the automorphism

$$\theta : (\operatorname{Spec} k[X])|_{D(X)} \xrightarrow{\sim} (\operatorname{Spec} k[X])|_{D(X)}$$

whose action on the Laurent ring is inversion. Define $\mathbb{P}^1_k = (\operatorname{Spec} k[X])^\theta$ by the preceding construction. Its two affine coordinates obey $X_1 = X_0^{-1}$ on the overlap. Proposition 12.1 and Theorem 9.6 give two affine chart restrictions covering it. Hence $\mathbb{P}^1_k$ is a qcqs scheme.

13.2. **Global functions.**

Lemma 13.1. *If $p, q \in k[X]$ satisfy $f(p) = \sigma(f(q))$ in L, then $p = q$ and both are constant polynomials.*

Proof. Use induction on a finite coefficient list representing q, with the statement quantified over all p. The conclusion is a property, so a mere coefficient representation suffices. If $q = 0$, injectivity of f gives $p = 0$.

Write $q = c + Xq'$. The overlap equation becomes

$$f(p) = f(c) + x^{-1}\tau(q').$$

For $p' = X(p - c)$ it follows that $f(p') = \tau(q') = \sigma(f(q'))$. By induction $p' = q'$ is constant. The constant coefficient of p' is zero, so $p' = q' = 0$. Regularity of X gives $p = c$, and consequently $q = c + Xq' = c$ as well. This argument needs neither a degree function nor a domain hypothesis. □

Theorem 13.2. *For every commutative ring k, the constant-section homomorphism*

$$k \longrightarrow \Gamma(\mathbb{P}^1_k), \qquad c \longmapsto (c|_\top, c|_\top)$$

is an isomorphism. Its inverse takes the constant coefficient of the polynomial on the first chart.

Proof. Identify global sections on each affine chart with $k[X]$. Under evaluation on the punctured chart, the overlap equation for a pair (p, q) is exactly

$$f(p) = \sigma(f(q)).$$

Indeed, the overlap automorphism is obtained from $\operatorname{Spec}\sigma$ through the principal-open isomorphism, so its action on sections is σ. Functoriality of global sections identifies the transition equation with the displayed one. Restriction between the equal overlap opens is invertible, so the implication works in both directions.

Every pair of identical constants satisfies the equation because σ fixes k. Lemma 13.1 shows conversely that every global section is such a pair. Reading the constant coefficient gives both inverse laws. The map is a ring homomorphism by the fibre-product universal property. □

13.3. **Non-affineness and the structure map.**

Theorem 13.3. *If $\mathbb{P}^1_k$ is affine, then $1 = 0$ in k. Hence it is non-affine whenever $1 \neq 0$ in k.*

Proof. The two components of every global section are the same constant polynomial, by Theorem 13.2. Its invertibility open is therefore diagonal, since the glued support is componentwise. Let

$$\pi : \mathrm{Zar}(\Gamma\mathbb{P}^1_k) \longrightarrow L_\theta$$

be the lattice map of the canonical affine morphism. Its generators map to invertibility opens of global sections, so their images are diagonal. Finite generation and preservation of joins imply that every image is diagonal. Here only diagonal pairs that belong to L_θ are used; their finite joins remain diagonal.

If the canonical map were an isomorphism, the chart $(\top, D_A(X))$ would lie in its image and hence be diagonal. Thus $D_A(X) = \top$. As in Theorem 11.4, the Nullstellensatz gives $1 = Xq$, and evaluation at zero gives $1 = 0$ in k. No identification of $\mathrm{Zar}(\Gamma\mathbb{P}^1_k)$ with $\mathrm{Zar}(k)$ is needed. □

Proposition 13.4. *There is a canonical local morphism*

$$\mathbb{P}^1_k \longrightarrow \operatorname{Spec} k$$

whose restriction to either affine chart is $\operatorname{Spec}(k \to k[X])$.

Proof. Apply Theorem 8.2 to the constant-section map $k \to \Gamma(\mathbb{P}^1_k)$ of Theorem 13.2. For either chart, compose with its inclusion and its isomorphism from $\operatorname{Spec} k[X]$. Its global component sends $c \in k$ to the global section of the constant polynomial c: restrict the constant pair to the chart, then take the corresponding component. This is also the global component of $\operatorname{Spec}(k \to k[X])$. Uniqueness in Theorem 8.2 proves the chart compatibilities. □

14. What has been formalised

The development in `unit/sag/qcqs` checks in Unit: the entry point `All.seg` reports 100 modules and no failures at the stage described here. Its parameters are univalence and replacement with the associated surjectivity principle. Function extensionality, propositional truncation and effective small set quotients are derived. The development uses none of the geometric axioms Loc, synthetic quasi-coherence or Zariski-local choice.

The main mathematical correspondence is as follows. File names refer to the directory above.

Mathematics	**Files and principal declarations**
Localisation and the elementary characterisation	`Localisation`, `LocalisationFractions`: `IsLoc`, `Elementary`, `Agree`
Radicals and the Zariski lattice	`Radical`, `FamilyLattice`, `ZariskiLattice`: `order`, `reflect`, `generated`, `presented`
Universal support	`DistLattice`, `ZariskiUniversal`: `factor`, `factor-D`, `unique`
Localisation bridge and restriction	`Bridge`, `Restriction`: `invertible-le`, `le-invertible`, `rho`
Section rings, basic opens and limits	`Sections`, `BasicOpen`, `Limit`: `Section`, `iso`, `natural`, `lift`
Basic separation and gluing	`Separation`, `NatBound`, `Gluing`: `injective`, `Cover.sheaf`
Sheaf on all finite covers	`Separated`, `RingedLattice`, `Sheaf`: `amalgam-prop`, `structure-sheaf`

Extension theorem	`Extension`: `extension-map`, `extend-restrict`, `extend-basic`, `Unique.agrees`
Invertibility opens and local structure	`InvOpen`, `SpecInv`, `LocallyRinged`, `SpecLocal`: `InvMap`, `spec`
Ring fibre products	`FibreProduct`: `Lift`, componentwise units
Doubling and the chart identifications	`DoubleLattice`, `Doubling`, `Chart`: `doubled`, `iota`, `to`, `natural`, `inv-to`, and the second-chart counterparts
Maps into a doubling	`Fold`, `IntoDoubling`, `DoublingUniversal`: `equivalence`, `local-equivalence`
Affine spectrum and doubled-origin line	`Affine`, `DoubledLine`: `spec`, `doubled-line`
Global sections of the affine spectrum	`Global`: `global`, `evaluate`, `iso`
The universal property of the affine spectrum	`SpecMap`, `SpecUnique`, `SpecUniversal`: `morphism`, `is-local`, `lat-eq`, `sharp-eq`, `component-of`
Restriction to an open	`Slice`: `restricted`, `inclusion`, `inclusion-local`
Isomorphisms and their locality	`Iso`: `IsIso`, `EqualOpens`, `Transfer`, `Local.local`
Chart isomorphisms onto restrictions	`ChartIso`: `chart1`, `iso1`, `local1`, and their second-chart counterparts
Composition and spectra of ring maps	`Compose`, `SpecHom`: `Identity`, `Then.component`, `Isomorphism.iso`
The canonical affine criterion	`AffineCriterion`: `IsAffine`, `Criterion.affine`
Principal affine opens	`SliceUniversal`, `SpecBasicOpen`: `Back`, `Restore`, `cancel`, `j-iso`, `k-iso`, `affine`
Qcqs schemehood and the doubled-origin example	`Qcqs`, `DoubledLineQcqs`: `AffineOpen`, `AffineCover`, `IsQcqs`, `affine1`, `affine2`, `qcqs`
Regularity and localisation at the variable	`PolynomialCoefficients` (shared from `sag/pn`), `VariableKernel`: `regular`, `Kernel.equal`, `one-zero`
Global sections of the doubled-origin line	`DoubledLineGlobal`: `separate`, `equal-components`, `diagonal`, `back`, `iso`
Non-affineness over a nonzero base	`DoubledLineNotAffine`: `d-diagonal`, `Image.diagonal`, `IfAffine.one-zero`, `not-affine`
Equality and sets of morphisms	`MapEquality`: `ext`, `local-ext`; `SpecUniversal.equivalence`
Doubled-line points and base change	`DoubledLinePoints`: `equivalence`; `DoubledLineBaseChange`: `natural`
Idempotents and complementary opens	`Idempotents`, `Interval`
Zariski lattice of a principal quotient	`ZariskiQuotient`; shared `PrincipalQuotient`, `PrincipalCertificates`
Idempotent point classification and naturality	`DoubledLineIdempotents`, `DoubledLineIdempotentsBaseChange`
Gluing with a transition and the projective line	`LatticePullback`, `GlueAlong`, `GlueTransition`, `ProjectiveLine`: `affine1`, `affine2`, `qcqs`
Inversion and the punctured chart	`Inversion`, `PuncturedChart`: `sigma`, `involution`, `automorphism`
Global functions on the projective line	`InversionOverlap`: `overlap`; `ProjectiveLineGlobal`: `diagonal`, `back`, `iso`
Non-affineness of the projective line	`ProjectiveLineNotAffine`: `not-affine`

Structure map and its chart restrictions	`ProjectiveLineStructure`: `structure`, `structure-local`, `First.compatible`, `Second.compatible`

The construction of the structure sheaf is parametric in the chosen principal localisations. `Affine` supplies the fraction rings as one such family. Restriction to a principal ideal is packaged as a locally ringed lattice, and the chart identifications are isomorphisms onto those restrictions. The affine criterion applies to any principal localisation family for A; the spectrum of ΓX used to define the canonical map uses the fraction-ring family. The formal affine criterion proves the substantive direction of Theorem 9.6; the other direction is the defining canonical isomorphism itself.

The universal-property proofs first compare lattice maps pointwise and section maps through restriction between equal target opens. `MapEquality` identifies this concrete equality with equality of the records, and proves that morphisms, and local morphisms, form sets. The inverse laws for the spectrum and doubling constructions are therefore exposed as actual equivalences. The spectrum statement remains parametric in the principal localisation family and explicitly assumes that the carrier of A is a set.

The doubled-line point classification is proved for k-algebras, with base change both in the extent formulation and in the idempotent formulation. The quotient-lattice proof uses universal supports; the complementary-open proof uses sheaf gluing. No synthetic geometric axiom is used in either.

For the projective line, the checked results include its construction as a locally ringed lattice, its two affine chart isomorphisms, qcqs schemehood, the canonical ring isomorphism with k, non-affineness when $1 \neq 0$ in k, and the explicit local structure map to $\operatorname{Spec} k$ with both chart compatibilities. The proof of Lemma 13.1 is the coefficient-list induction used in the formal development.

Not yet formalised at this checkpoint are a general mapping property for gluing with a transition, the classification of maps into the projective line by extents and reciprocal sections, and its comparison with rank-one direct summands of B^2. No proof of non-separatedness of the doubled-origin line or separatedness of the projective line is claimed here. Category-theoretic packaging and comparison with functorial schemes are outside the present account.

References

[1] Thierry Coquand and Henri Lombardi, *Hidden constructions in abstract algebra: Krull dimension of distributive lattices and commutative rings*, in *Commutative Ring Theory and Applications*, Lecture Notes in Pure and Applied Mathematics **231**, Marcel Dekker, New York, 2003, pp. 477–499. Updated version: arXiv:1801.00097.

[2] Thierry Coquand, Henri Lombardi and Peter Schuster, *The projective spectrum as a distributive lattice*, Cahiers de Topologie et Géométrie Différentielle Catégoriques **48** (2007), no. 3, 220–228. Numdam.

[3] Thierry Coquand, Henri Lombardi and Peter Schuster, *Spectral schemes as ringed lattices*, Annals of Mathematics and Artificial Intelligence **56** (2009), nos. 3–4, 339–360. doi:10.1007/s10472-009-9160-7.

[4] Max Zeuner, *Univalent Foundations of Constructive Algebraic Geometry*. Source copy: `SCHEME/zeuner1.pdf`.

[5] Max Zeuner and Anders Mörtberg, *A Univalent Formalization of Constructive Affine Schemes*, LIPIcs TYPES 2022, article 14, doi:10.4230/LIPIcs.TYPES.2022.14. Source copy: `SCHEME/zeuner2.pdf`.

[6] Valery Isaev, *A Simple and Constructive Approach to Schemes*. Source copy: `SCHEME/isaev.pdf`.